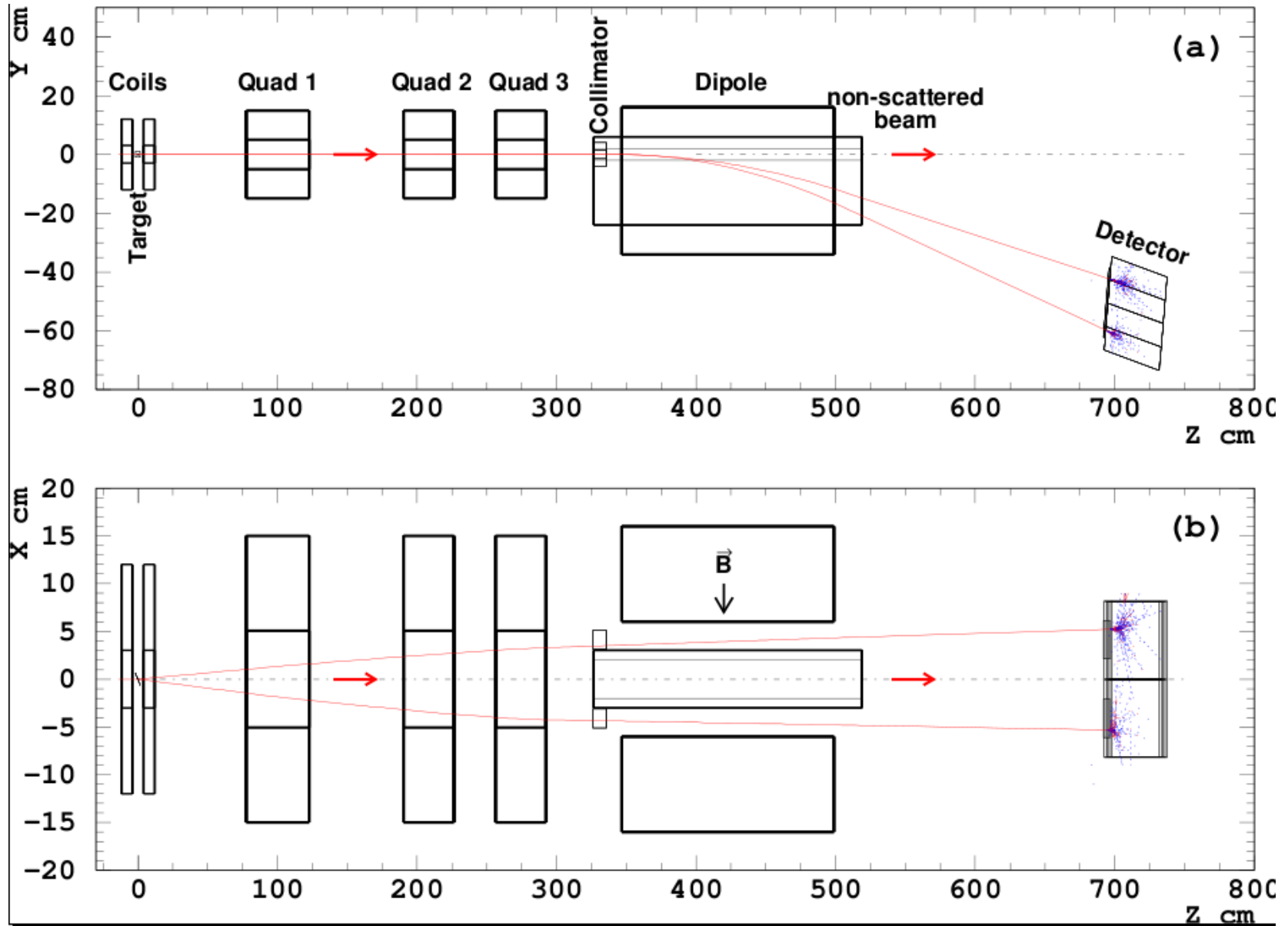


Hall A Møller Polarimeter after Upgrade

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Møller Polarimeter Layout



Why We Need Upgrade?

Møller polarimeter now:

- *the beam current $0.3 \div 1 \mu\text{A}$;*
- *closed slit (slit position from 14 to 18 r.u.);*
- *no energy/position fast feed-back*

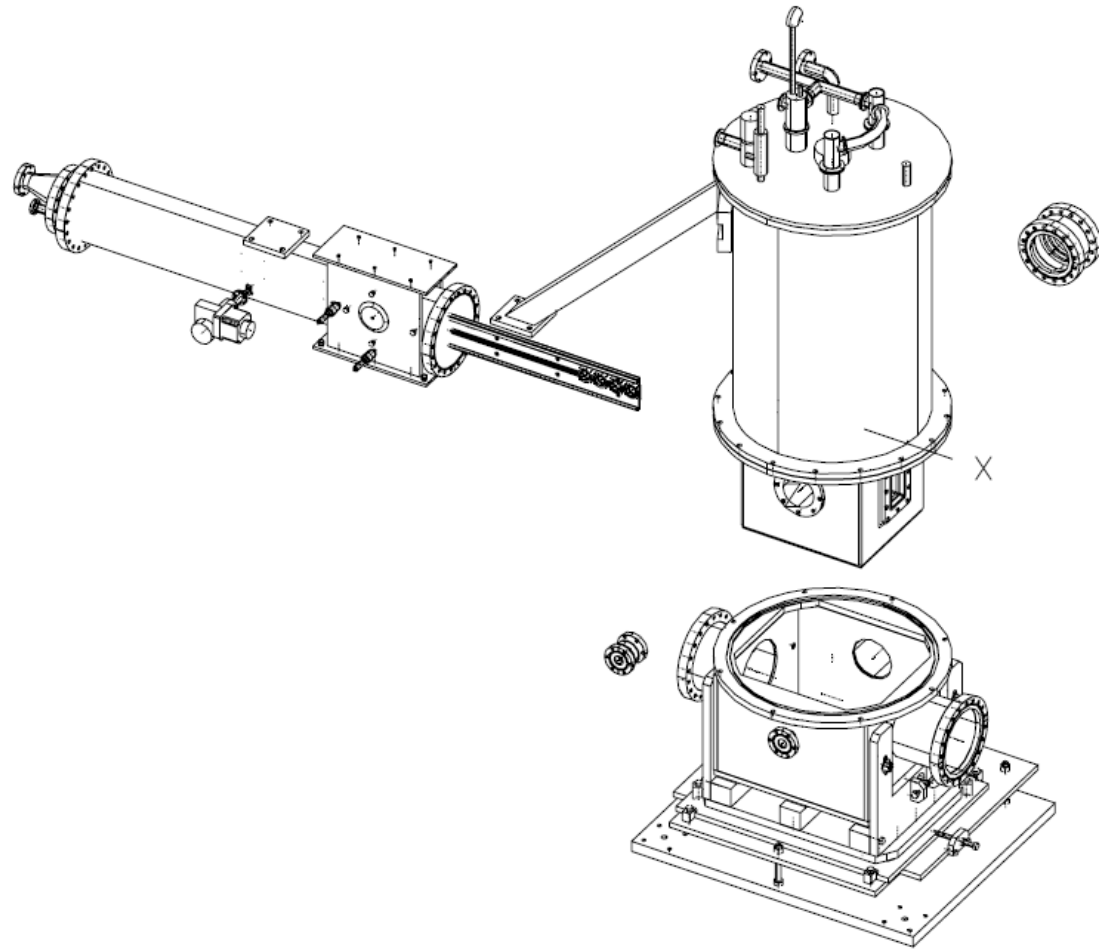
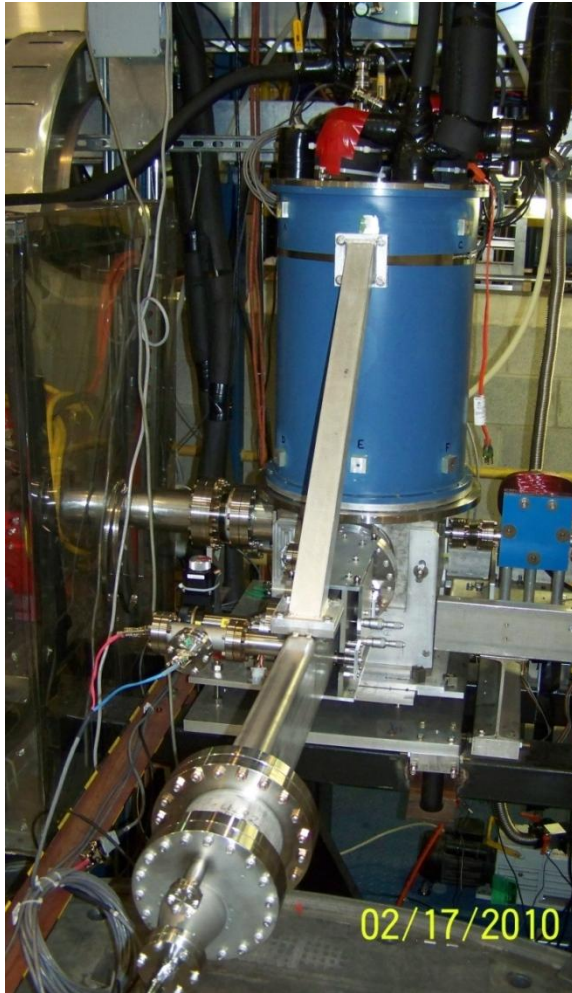
- 1. Polarized electron target "brute force" 3T (100% target magnetization);*
- 2. Introduce a beam duty cycle:
reduce target heating, increase beam current:*
 - *no laser phase dependence (fully open slit);*
 - *energy fast feed-back;*
 - *beam position fast feed-back;*
- 3. New fast DAQ based on FADC (higher rate)*
- 4. Segmented aperture detector (higher rate)*

Møller Polarimeter Systematic Error

Variable	Hall A	
	<i>Present</i>	<i>Upgrade</i>
<i>Target polarization</i>	1.8%	0.5%
<i>Target angle</i>	0.5%	0.0%
<i>Analyzing power</i>	0.3%	0.3%
<i>Levchuk effect</i>	0.2%	0.2%
<i>Target temperature</i>	0.0%	0.02%
<i>Dead time</i>	0.3%	0.3%
<i>Background</i>	0.3%	0.3%
<i>Others</i>	0.3%	0.5%
<i>Total</i>	2.0%	~1.0%

10 μ A - "the same" beam condition as running experiment

New Møller Target Design

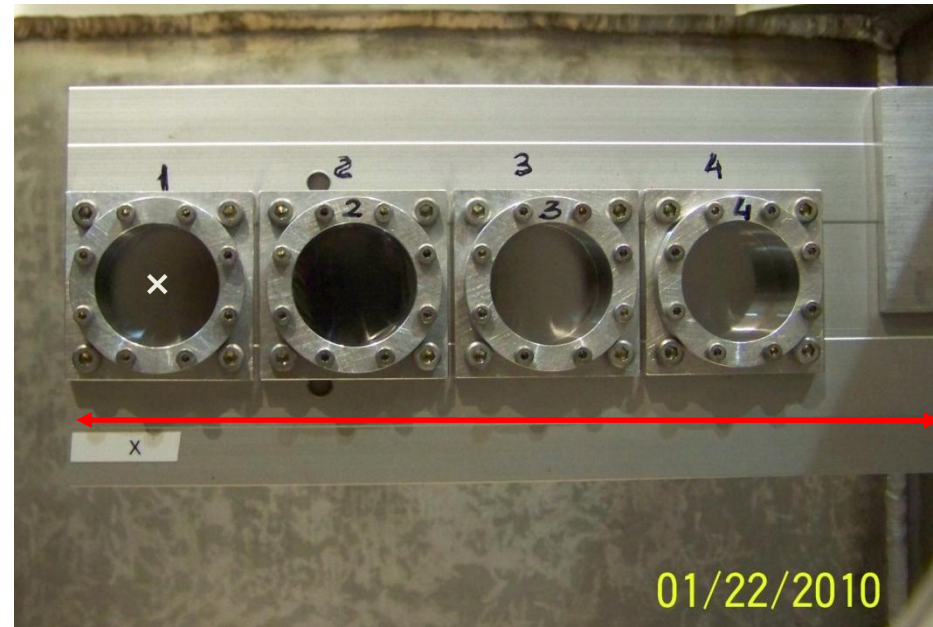
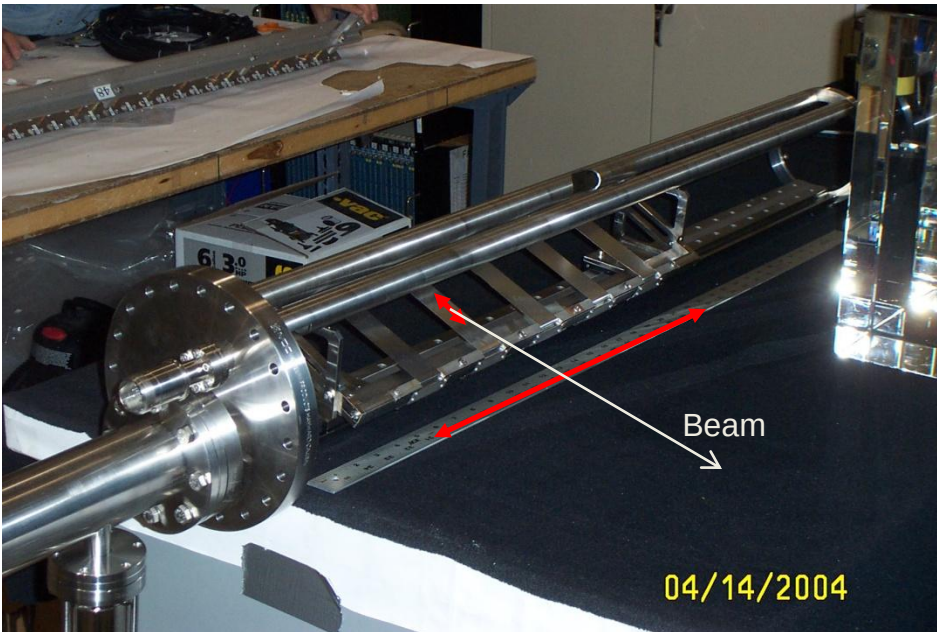


*Superconducting Helmholtz Coils ~3 T magnetic field
For PREX: ~7° of the beam XY rotation
~45 min to rump up/down*

Møller Target Holder

Old Target Holder

New Target Holder



5 target foils:

SM-7 μ m, Fe-7 μ m, Fe-14 μ m,

SM-29 μ m, SM-13 μ m

SM- Supermendur (49%Fe 49% Co2%V)

Tilted 20° to the beam direction

Moved along the beam and

in transverse to the beam direction

Can be moved with the beam ON

4 target foils:

Fe-1 μ m, Fe-1 μ m, Fe-4 μ m, Fe-12 μ m

Tilted 90° to the beam direction

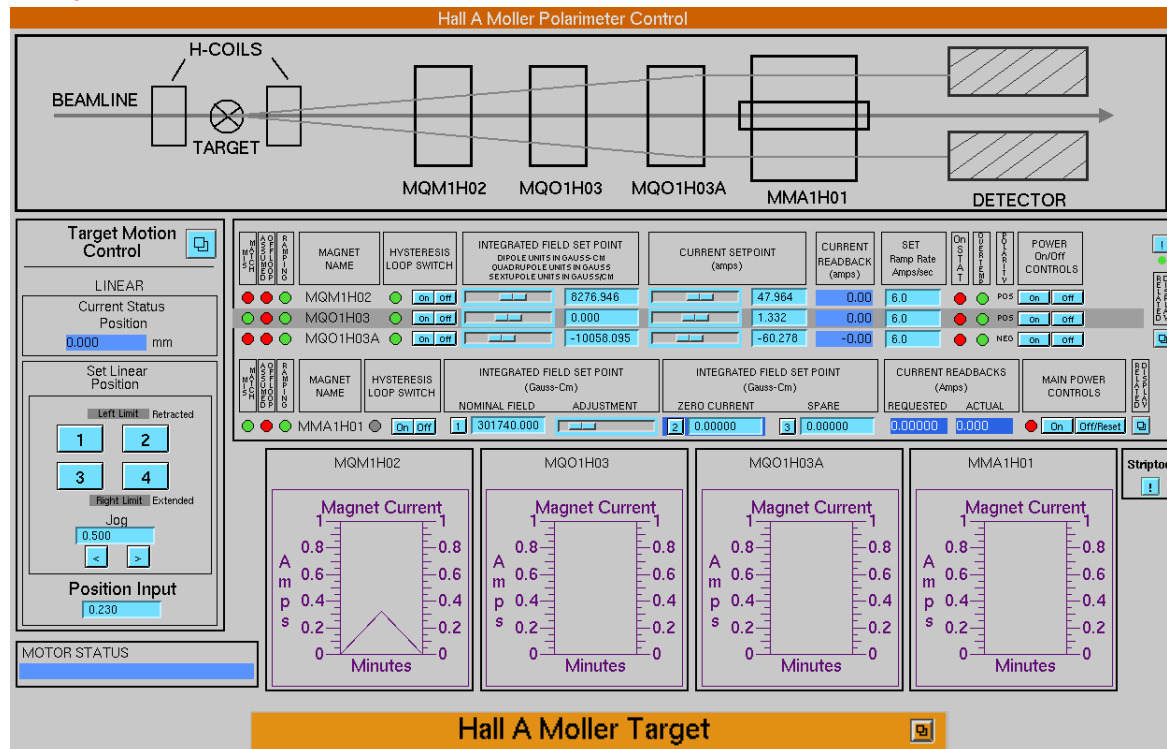
Moved in transverse to the beam

direction

The beam **MUST BE OFF** for any target movement

Only MCC can move the target

Updated MCC Møller screens



Møller Requirements for the Beam

- *Fast Raster $1 \times 0.5 \text{mm}^2$ (has to be changed for Møller measurements);*
- *Beam position:
centered $XY \pm 0.2 \text{mm}$ (BPM1HO1);
parallel (the best tuning);*
- *Measurements with **CW** beam up to $5 \mu\text{A}$;*
- ***USER mode**
beam current up to $15 \mu\text{A}$;
fully open slit (use attenuator for current);*
- ***Beat Frequency mode** (for special test):
beam current up to $50 \mu\text{A}$;
micro-suppression $n=10$ (50MHz)*

New Møller FSD

Fast Shutdown System - Moeller Target Masking

The Question: Should we mask the Hall A Moeller Target FSD?

Target Mask Request Request Readback

Target UNMask Request

-1.00 Slit Value iocin1:heartbeat 182336

0.00 BLA Current iocbcm:heartbeat 706975

ISD1H001L.H
(Moeller Mask Request)

ISD1H001NA.B7
(Moeller Mask Hardware Readback)

The Judges

Judge	Status
iocin1 Heartbeat	Yea
iocbcm Heartbeat	Yea
Slit A Status	Nay
Hall A BCM Status	Yea
All Sequence Channels Connected	Yea

The Decision

Masking OFF!

Application is fsdwft (wft = waterfall target) -- also see the fsd application.

The Moeller Target can be moved without an FSD trip, only if the Moeller FSD is masked.
The Moeller FSD can be masked only from this screen via the above [0] Target Mask Request button.
Masking is allowed only when all 5 Judges are green (Yea).
The Slit A Judge is green only for a slit position between 8 and 30.
The Hall A BCM Judge is green only for a beam current of less than 5 uA.

CW mode - FSD the same

USER mode - slit fully open, beam current limit 60μA

Hall A will never request masking of Møller target movement

Time Budget for Møller Measurement

Møller settings (+magnet ramp up): ~1hour

Raster $1 \times 0.5 \text{mm}^2$

CW: (low current $< 5 \mu\text{A}$)

$1 \mu\text{m} / 3 \mu\text{A} \sim 1 \text{hour}$ (0.3%stat error)

$4 \mu\text{m} / 3 \mu\text{A} \sim 20 \text{min}$ (0.3%stat error)

$12 \mu\text{m} / 1 \mu\text{A} \sim 20 \text{min}$ (0.3%stat error)

Target thickness and condition systematic

CW: ~2hours

"User mode": (high current $> 5 \mu\text{A}$)

10ms of 30ms helicity pulse (30Hz)

$1 \mu\text{m} / 10 \mu\text{A}$ beam: $\sim 45 \text{min}$ for 0.3% stat.er.

"User mode": ~1hour

Restore PREX beam settings: ~1hour

Total: ~5hours